

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN

COURSE CODE: CSA1142

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing techniques. (BL4,BL5) Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:40

Annexure A

MOCK INTERVIEW

Code of Conduct:

I K Madan Mohan Reddy (192411206)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Byreddy Madan

RUBRICS FOR CALCULATION:

MOCK INTERVIEW

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problemsolving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

Q2. Scenario: Food Delivery Application Features:
Browse Menu, Place Order, Payment

Question:

Compare **functional testing** and **usability testing** in this system. When should each be applied, and why are both important?

Scenario: Food Delivery Application

Features:

- Browse Menu
- Place Order
- Payment

Compare Functional Testing and Usability Testing in a Food Delivery Application

Food delivery applications have become an essential part of modern life. Users expect these applications to be accurate, reliable, and easy to use. To ensure high-quality software, different types of testing are performed during software development. Among them, Functional Testing and Usability Testing are two important testing methods. Although both aim to improve the quality of the application, they focus on different aspects of the system.

Introduction

A food delivery application allows customers to browse menus, select food items, place orders, make payments, and track deliveries. If any function does not work correctly, users may be unable to order food. Similarly, if the application is difficult to use, customers may become frustrated and switch to competitors.

Therefore, both functional testing and usability testing are necessary to ensure that the application performs correctly and provides a positive user experience.

Functional Testing

Definition

Functional testing is a software testing technique that verifies whether the application functions according to the specified requirements. It checks the behavior of each feature and ensures that the system produces the expected output for a given input.

The primary objective of functional testing is to validate the business logic and functionality of the application.

Functional Testing in a Food Delivery Application

1. Browse Menu Feature

The tester verifies:

- Categories are displayed correctly.
- Food items appear under the correct category.
- Search functionality works properly.
- Filters such as vegetarian, non-vegetarian, and price range work correctly.
- Food descriptions, prices, and images are displayed accurately.

Example Test Case

Input: Search for "Pizza"

Expected Output: All pizza items should be displayed.

Actual Result: Compare actual results with expected results.

2. Place Order Feature

The tester verifies:

- Items can be added to the cart.
- Quantity can be increased or decreased.
- Total amount is calculated correctly.
- Delivery address can be selected.
- Order confirmation is generated successfully.

Example Test Case

Input: Add 2 burgers costing ₹150 each.

Expected Output: Total should be ₹300.

3. Payment Feature

The tester verifies:

- Credit card payments work correctly.
- Debit card payments work correctly.
- UPI payments are processed successfully.
- Wallet payments function properly.
- Payment confirmation messages are displayed.
- Failed transactions are handled correctly.

Example Test Case

Input: Make payment using UPI.

Expected Output: Payment should be successful and order should be confirmed.

Advantages of Functional Testing

1. Ensures all features work correctly.
2. Detects defects in business logic.
3. Validates system requirements.
4. Improves software reliability.
5. Reduces production failures.

Limitations of Functional Testing

1. Does not evaluate user satisfaction.
2. Cannot determine whether the interface is easy to use.
3. Focuses only on functionality.
4. May miss user experience problems.

Usability Testing

Definition

Usability testing is a software testing technique that evaluates how easy and user-friendly an application is for end users. It focuses on user experience, interface design, navigation, efficiency, and satisfaction.

The main objective is to determine whether users can perform tasks easily without confusion.

Usability Testing in a Food Delivery Application

1. Browse Menu Feature

Usability testers observe:

- Whether users can find food items quickly.
- Whether categories are easy to understand.

- Whether search options are visible.
- Whether menu layout is attractive and organized.
- Whether images help users make decisions.

Example

A user wants to order a pizza.

Questions:

- Can the user find the pizza section easily?
- Is the search bar visible?
- Does the user get confused while browsing?

2. Place Order Feature

Usability testers evaluate:

- Ease of adding items to the cart.
- Simplicity of quantity adjustment.
- Visibility of cart information.
- Ease of selecting delivery address.
- Clarity of order summary.

Example

The user should be able to place an order within a few steps without requiring instructions.

3. Payment Feature

Usability testers observe:

- Whether payment options are easy to locate.
- Whether payment instructions are clear.

- Whether error messages are understandable.
- Whether confirmation screens provide sufficient information.

Example

After payment, users should clearly understand whether the transaction was successful or failed.

Advantages of Usability Testing

1. Improves user satisfaction.
2. Enhances application usability.
3. Reduces user frustration.
4. Improves navigation and accessibility.
5. Increases customer retention.

Limitations of Usability Testing

1. Does not verify business logic.
2. Requires participation of real users.
3. Can be time-consuming.
4. Results may vary among users.

When Should Functional Testing Be Applied?

Functional testing should be performed:

During Development

Developers and testers verify each module individually.

After Feature Completion

When browsing, ordering, and payment modules are developed.

Before Integration

To ensure each component functions properly.

During System Testing

To verify complete application functionality.

Before Release

To ensure all business requirements are satisfied.

Why?

Because incorrect functionality can lead to:

- Wrong order placement
- Payment failures
- Incorrect billing
- Customer complaints
- Financial losses

When Should Usability Testing Be Applied?

Usability testing should be performed:

During UI/UX Design

To evaluate initial screen designs.

After Prototype Development

To collect user feedback.

Before Product Launch

To ensure users can navigate easily.

After Major Updates

To verify that new features remain user-friendly.

Why?

Because poor usability can result in:

- User confusion
- Increased abandonment rate
- Negative reviews
- Reduced customer loyalty
- Lower business revenue

Why Both Functional Testing and Usability Testing Are Important

A food delivery application must not only work correctly but also provide a smooth user experience.

Consider the following scenarios:

Scenario 1

The payment gateway processes payments correctly.

Functional Testing Result:

✓ Pass

However, the payment button is difficult to find.

Usability Testing Result:

✗ Fail

Users may leave without completing the payment.

Scenario 2

The search function is easy to locate.

Usability Testing Result:

✓ Pass

But the search returns incorrect results.

Functional Testing Result:

✗ Fail

Users cannot find desired food items.

Scenario 3

The order placement process works correctly.

Functional Testing Result:

✓ Pass

But users must go through ten complicated screens.

Usability Testing Result:

✗ Fail

Users may uninstall the application.

Benefits of Using Both Testing Methods Together

1. Ensures accurate functionality.
2. Improves customer satisfaction.
3. Reduces application errors.
4. Enhances user experience.
5. Increases customer trust.
6. Improves application performance.
7. Reduces maintenance costs.
8. Increases business revenue.
9. Encourages repeat usage.
10. Strengthens brand reputation.

Conclusion

Functional testing and usability testing are both essential for the success of a food delivery application. Functional testing verifies that features such as browsing menus, placing orders, and making payments work according to requirements. Usability testing ensures that these features are simple, intuitive, and convenient for users.

